

Power BI Modeling

30-Minute Hands-On Exercise | Week 9 - Section 2B

Learning Objectives: Explain the structure of a Power BI semantic model | Distinguish data tables from lookup tables and apply correct naming conventions | Describe how relationships, cardinality, and granularity affect model accuracy | Configure column properties including data type, formatting, sort order, data category, default summarization, and navigational hierarchies

Instructions: Complete all four parts in 30 minutes. Refer to your Section 2B notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 2B.

1.	In Power BI, a data model is referred to as a _____ .
2.	Tables that contain numeric measures and represent events (the center of a star schema) are called _____ .
3.	Tables that contain descriptive attributes used to group or filter facts (the outer tables of a star schema) are called _____ .
4.	Relationships between data tables and lookup tables should have _____ cardinality -- many records on the data side, one record on the lookup side.
5.	The level of detail at which data is stored in a table -- for example, one row per transaction vs. one row per day -- is called _____ .
6.	To make a Month Name column sort correctly (January before July), you configure a _____ that points to a Month Number column.

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each term.

Term	Description
A. Semantic Model	1. Synonymous with a fact table; records describe events and contain measures
B. Cardinality	2. A metadata tag on a column that enables specialized visualization behavior (e.g., mapping)
C. Navigational Hierarchy	3. A drill-down order added to visualizations, such as Year > Quarter > Month > Day

D. Data Table	4. The aggregation type Power BI applies to a column when it is dragged into a visual (e.g., Sum)
E. Default Summarization	5. Power BI's term for a data model; built on the xVelocity engine
F. Data Category	6. The explicitly defined numerical relationship between two sides of a model relationship (e.g., many-to-one)

Part 3: Column Properties (~8 minutes)

For each situation below, identify which column property you would configure. Choose from: Data Type | Numeric Formatting | Date/Time Formatting | Data Category | Sort By Column | Default Summarization | Navigational Hierarchy.

#	Situation	Column Property to Configure
1	A city name column is not rendering on a map visual even though the data looks correct.	
2	A 'Popularity Rating' column (values 0-5) is being summed by default, which produces meaningless totals.	
3	A 'Month Name' column sorts alphabetically (April, August, December...) instead of chronologically (January, February, March...).	
4	A column imported from a CSV is stored as text but contains date values that need to support time-intelligence calculations.	
5	A revenue column needs to display values as \$1,200.00 with a dollar sign and two decimal places.	
6	You want report users to be able to click into a map visual and drill from Country down to Region, then City.	

Part 4: Scenario Analysis (~9 minutes)

Read each scenario and apply a concept from Section 2B. Write the relevant concept and a brief explanation in the right column.

Scenario	Your Answer + Brief Explanation
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1. A developer names their tables 'FactSales' and 'DimCustomer' in the Power BI semantic model. A colleague says this is a problem. What naming convention should be used instead, and why?	
2. An analyst joins a daily budget summary table directly to a transaction-level sales table. Revenue totals suddenly look 30x higher than expected. What modeling concept explains this problem?	
3. A data model has a relationship drawn directly between two data (fact) tables -- an Orders table and a Returns table. What schema rule does this violate, and how should it be fixed?	
4. A report displays a column called 'Customer Name Amount.' A stakeholder is confused by the label. Using Section 2B naming guidelines, identify at least two problems and suggest better names.	
5. A product manager wants to click on a bar chart and drill from Category down to Size and then Color. What Power BI feature would you configure to enable this, and what would the hierarchy look like?	

Bonus / Reflection (if time allows)

Section 2B states: 'Most difficult data analysis problems are rooted in a flawed model.' In 3-5 sentences, explain what this means using a real or hypothetical example. What kinds of report errors or incorrect numbers could a poor model cause, and how does following star/snowflake schema rules help prevent them?